Afghanistan Islamic Emirate

Ministry of Higher Education

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RANA University

Directorate of Computer Science Faculty

Software Engineering Departement

امارت اسلامی افغانستان

وزارت تحصیلات عالی

ریاست موسسات خصوصی تحصیلت عالی

پوهنتون رنا

ریاست پوهنحی کمپیوتر ساینس

آمریت دیپارتمنت انجنیری نرم افزار

MONOGRAPH

ON

Rana University Library Management System

سیستم مدیریت کتابخانه پوهنتون رنا

BY **Elham Isaac**

18-RV200-315

In partial fulfillment of the requirements for the award of the degree of

BACHELOR OF SOFTWARE ENGINEERING

BSE

TO

RANA University Baraki Square, Kabul–Afghanistan

Afghanistan Islamic Emirate
Ministry of Higher Education
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RANA UNIVERSITY

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ACKNOWLEDGMENT

All praises and thanks to Almighty Allah, the source of knowledge and wisdom to mankind, who conferred me with power of mind and capability to take this material contribution to already existing knowledge. All respect and love to him who is an everlasting model of guidance for humanity.

I would like to express the deepest appreciation to my colleagues who encouraged me in writing my monograph on “Rana University Library Management System” with the attitude and the substance of a genius, they continually and convincingly conveyed a spirit of adventure accordingly. I wish to thank my project supervisor **Mr. Khwaja Hameedullah Shiam** whose guidance made my project possible. His encouragement and wisdom made my efforts worthwhile. It is with great honor that I would also like to thank my family and friends, whose names I have not mentioned, how yet supported and helped me in one way or the other. Finally, I thank you, the reader for taking time to read my thesis.

Signature

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Chapter No 1

INTRODUCTION

1.1 Introduction

The **Rana University Library Management System** is a desktop-based software application developed using **Java Swing (NetBeans IDE)** and **MySQL (XAMPP)** as the backend. It is designed to digitize and streamline the daily operations of the university's library, such as book and student management, issuing and returning books, overdue tracking, and report generation.

The goal of this system is to reduce manual paperwork, improve data accuracy, and enhance operational efficiency for librarians and staff. The system features a user-friendly graphical interface, with a focus on real-time data visualization and automation of critical tasks. It eliminates the traditional pen-and-paper approach and introduces dynamic data handling, interactive dashboards, and secure user management.

The home page provides quick access to essential statistics such as total books, students, issued books, and overdue records. Additionally, a pie chart graphically represents the proportion of issued books, offering staff a visual summary of library activity. The system includes a secure login/signup mechanism, and role-based access, ensuring only authorized personnel can manage records.

1.2 Overview

The system is built using a **structured, modular, and menu-driven architecture**, allowing easy navigation and future scalability. Designed with simplicity in mind, the software ensures that users — even those with minimal technical expertise — can operate it effectively.

Key features include:

- Book and student records management

- Issuing and returning books
- Overdue book tracking (defaulter list)
- Report generation based on custom date ranges
- User authentication and account creation
- Graphical data representation (Pie Chart)

This system enhances data accessibility, improves decision-making for administrators, and reduces operational overhead. With centralized data storage, all actions are recorded and retrievable, thus improving transparency and accountability in library operations.

1.3 Objectives

The main objectives of the Rana University Library Management System are:

- To create a centralized system that automates routine library processes
- To minimize manual work and improve operational speed
- To securely manage student and book data
- To monitor overdue books and maintain an up-to-date defaulter list
- To generate accurate reports for library activity
- To offer an intuitive and user-friendly interface
- To enable easy modifications and scalability for future enhancements

1.4 Project Profile

Project Title	**Rana University Library Management System**
Platform	Desktop Application

Operating System	Windows 7/8/10, Linux
Front-End Technology	Java (Swing Framework) using NetBeans IDE
Back-End Technology	MySQL (XAMPP)
Database Management	Structured Query Language (SQL)
Reporting/Visualization	Java-based Pie Chart Integration **Tool**

1.5 Purpose

The purpose of this project is to develop a **user-friendly, interactive, and reliable system** that digitizes the entire library operation of Rana University. It addresses core issues found in traditional systems, such as time-consuming manual processes, data loss, and the inability to quickly retrieve or modify records.

Primary goals include:

- Simplify the management of books and students
- Ensure faster transactions with minimal human error
- Maintain historical data for reporting and auditing
- Support efficient decision-making with accurate statistics
- Enhance security and restrict unauthorized data access

1.6 Benefits

The implementation of this system provides numerous benefits:

1. **Data Centralization:** All book, student, and transaction data is stored in a central database, ensuring accessibility and consistency.
2. **Automation:** Manual operations such as record-keeping and overdue tracking are fully automated.
3. **Real-time Monitoring:** Staff can view real-time statistics and reports.
4. **Reduced Redundancy:** Eliminates duplication of data and ensures accuracy.
5. **Improved Efficiency:** Reduces the time and manpower needed for library management.
6. **Security:** Role-based access control restricts unauthorized actions.
7. **Data Preservation:** Historical records can be maintained indefinitely without degradation.

1.7 Scope

This system is developed specifically for the Rana University Library. It manages all major operations such as:

- Student and book registration
- Book issuance and return
- Defaulter list management
- Visualization of data using pie charts
- Report generation for specific time periods
- User authentication and secure access control

Future scalability is supported, allowing integration of advanced modules like barcode scanning, notifications for overdue books, and online student access.

Out of Scope

- Integration with other university systems (e.g., finance or HR)
- Online book reservations and payments
- Mobile application support (currently desktop only)

1.8 Theoretical Background

The traditional library system in many universities still relies heavily on manual processes. This includes physical registers for student and book information, handwritten records of issued and returned books, and time-consuming report preparation.

Key problems in the manual system:

- High chance of human error
- Delays in retrieving or updating records
- Lack of data security
- Inability to generate instant reports
- Redundancy and inconsistencies in data

The Rana University Library Management System resolves these problems by introducing a digital environment where all data is stored, processed, and accessed through an automated interface.

1.9 Technology Used

1.9.1 Java (Swing Framework)

Java is a high-level, object-oriented language widely used for building desktop applications. Swing is part of Java Foundation Classes (JFC) and provides a rich set of GUI components for developing cross-platform applications. NetBeans IDE offers tools such as visual UI designers and code editors which simplify development.

1.9.2 MySQL (XAMPP)

MySQL is a powerful, open-source RDBMS used for data storage and retrieval. XAMPP simplifies database setup and includes Apache, MySQL, and PHPMyAdmin.

1.9.3 Chart Libraries

Java charting libraries (e.g., JFreeChart or JavaFX PieChart) are used to visualize data trends, such as the number of books issued over time.

1.10 Admin Characteristics

The system administrator is expected to:

- Be proficient in basic computer operations
- Understand the library workflow
- Be capable of maintaining user accounts and managing book/student records
- Monitor overdue book reports and issue actions accordingly

1.11 Challenges & Limitations

Despite its capabilities, the system faces the following limitations:

1.11.1 Time-Consuming Setup

Initial data entry (e.g., adding all existing students and books) may require considerable time.

1.11.2 Data Preservation

While the system maintains digital records, regular database backups are essential to avoid data loss from technical failures.

1.11.3 Cost of Hardware

Deploying the system across multiple library sections may require investment in computers and networking.

1.11.4 Change Management

Training sessions may be necessary to help staff transition from manual to digital systems.

1.11.5 Security

Although the system provides authentication, physical access to the computer system must be controlled to ensure complete data security.

1.11.6 Human Error

Incorrect data entries during manual input can still affect the system if proper validation is not enforced.

Chapter No 2

EXISTING COMPUTERIZED SYSTEM

2.1 Overview of the Current System

Rana University currently lacks a fully functional and reliable computerized library management system. Although some computerization efforts have been made recently, the existing system remains partially implemented and operational only in limited areas. This has led to multiple issues that hinder efficient library operations.

The current system still relies heavily on manual processes, which creates serious challenges in maintaining data integrity, ensuring security, and achieving smooth communication between departments. Additionally, the limited computerized modules suffer from vulnerabilities such as virus threats, lack of robust access control, and slow data processing. As a result, the system falls short in delivering the speed, efficiency, and reliability expected from a modern library management system.

2.2 Existing Manual System

Despite the partial digitization, most library operations at Rana University are still handled manually. This includes book issuance, student record-keeping, and inventory management.
The absence of a centralized system creates multiple issues such as:

- Data loss or duplication due to human error
- Poor tracking of books and student activity
- No streamlined control over staff and library resources
- Difficulty in retrieving historical data and statistics
- High dependency on clerical labor • Inefficient use of time and physical space **Drawbacks of the Manual System:**

- Time-consuming clerical work
- Poor maintainability and accuracy of records
- Limited accessibility to past records
- Delays in data search and retrieval
- Requirement of excessive human resources

- Inefficiency in space usage for physical records
- Risk of paper damage and long-term storage issues

2.3 Proposed System

The proposed **Rana University Library Management System** aims to eliminate the limitations of the current manual and partially computerized processes. Developed using **Java (NetBeans)** as the front-end and **MySQL** as the back-end, this system introduces automation, validation, and real-time data access for all library activities.

This solution provides:

- Centralized book and student management
- Secure access to book issuance and return records
- Accurate validation to prevent data errors
- High-speed processing with minimal memory consumption
- Secure login and administrative control

Key Features of the Proposed System:

- Strong data security and user authentication
- User input validation to ensure data correctness
- Efficient memory usage and fast performance
- Ordered and well-formatted data reporting
- Adaptability to modern technological standards

2.3.1 Benefits of the Proposed System

2.3.1.1 Standard Computerized Representation

The new system offers a unified and standardized digital platform, making data handling more reliable and less effort intensive.

2.3.1.2 Complete Information Management

All student and book-related information is stored digitally and accessible at the click of a button.

2.3.1.3 Faster Data Retrieval

Searching for student or book records is quick and efficient by using search fields like ID or name.

2.3.1.4 Reduced Workforce Requirement

Automation reduces the need for extensive staff while improving productivity.

2.3.2 Need for the Rana University Library Management System

In the digital age, institutions must adapt to modern technology to remain efficient, costeffective, and competitive. The Rana University Library Management System supports this transition by:

- Reducing reliance on paper-based documentation
- Facilitating centralized book and user data storage
- Improving communication and reducing data redundancy
- Allowing future integration with partner libraries for shared inventory access

2.3.3 Background and Motivation

Traditionally, library management relied on handwritten records and ledgers, requiring extensive manual effort. Over time, the need for faster, more accurate systems gave rise to computerized inventory and management tools. Rana University's need for a secure, centralized, and easy-to-maintain system was the driving force behind this project. The proposed system aims to minimize data loss, improve operational efficiency, and reduce costs associated with manual record-keeping.

2.3.4 Target Users and Market

The proposed system is designed primarily for small- to medium-sized institutions that require an affordable and user-friendly library management system. It avoids complex server requirements and can run efficiently on personal computers or laptops. Although currently developed for a single-user (admin-only) environment, it is scalable for future enhancements.

2.3.5 Transition from Paper to Computer-Based System

Modern digital systems have revolutionized how data is stored, retrieved, and managed. With computerization:

- Inventory is tracked automatically
- Students can be contacted efficiently
- Reports are generated in real time
- Data can be preserved securely for years

Compared to historical paper-based systems, the computerized model is faster, safer, and more reliable. The Rana University Library Management System brings this transformation to the university's library operations.

2.3.6 Role of Point-of-Sale (POS) Systems in Evolution

Inspired by inventory tracking models such as POS systems used in retail, the proposed system adopts a similar digital framework tailored for library operations. POS systems replaced mechanical registers, enhancing efficiency, especially in small businesses. Likewise, this project brings structured inventory and transaction logging (issue/return) to library services, minimizing errors and delays.

2.3.7 Importance of Computerization

Computerization brings:

- Improved accuracy and speed
- Lower operational costs
- Better decision-making due to reliable data
- Reduced manpower requirement
- Enhanced user satisfaction and experience

By shifting from manual to digital, the Rana University Library will modernize its services, matching current technological standards and future-ready infrastructures.

2.3.8 Survey of Current Inventory Systems

A recent survey revealed that:

- 23% of small businesses use no inventory system
- 30% rely on pen and paper
- 32% use general tools like Excel

This highlights a large untapped market for cost-effective inventory solutions. The Rana University Library Management System can serve as a model for transitioning to digital systems within limited budgets.

2.3.9 Project Requirements

The system is designed to meet the following core needs:

- Real-time book issuance and return tracking
- Centralized student and book data storage
- Customizable reports and charts
- Secure access control for the admin
- Compatibility with modern PCs and laptops
- Scalability for future enhancements

This ensures a robust and comprehensive solution tailored to the needs of Rana University and similar institutions.

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Chapter No 3
REQUIREMENTS GATHERING

Requirements gathering is one of the most essential phases in the development lifecycle of any software project. The success or failure of a project often hinges on how effectively requirements are captured, validated, and agreed upon with stakeholders. For the *Rana University Library Management System*, gathering precise requirements was especially critical to ensure the new system resolved the inefficiencies of the existing manual setup and met the specific needs of administrators managing books, students, and library operations.

3.1 Statement of Requirements

The statement of requirements defines the scope, environment, and essential functionalities of the system. It serves as a formal agreement between the developer and the stakeholders.

Key contents include:

- A concise requirement specification for managerial review.
- A list of core objectives and functionalities of the system.
- Details about the operating environment.
- Constraints related to design and implementation.
- References to relevant documentation and current system challenges.

This document was validated and approved by the university librarian and the IT supervisor, establishing a mutual understanding of the system's deliverables.

3.2 Rules for Successful Requirements Gathering

The following principles guided the requirement gathering process for this project:

- Never assume — always ask the user what they need.
- Engage users (admin/librarian) from the beginning.
- Define and agree on the scope early.
- Ensure requirements are **SMART**: Specific, Measurable, Agreed, Realistic, and Timebound.
- Always clarify doubts to avoid misinterpretation.
- Document requirements clearly and review them with users.
- Avoid jumping into solutions before understanding the problem fully.
- Use visual aids or prototypes to refine understanding.
- Finalize the requirements before system development begins.

3.3 Common Mistakes in Requirements Gathering

To maintain the integrity of the requirements process, several pitfalls were actively avoided:

- Designing solutions based on overly complex or experimental technologies not suited for real use.
- Neglecting to consult real users or the librarian responsible for library operations.
- Attempting to solve problems before fully understanding them.

3.4 Non-Functional Requirements

These describe the quality attributes of the system rather than specific behaviors:

Availability

- The system must be available during university hours without delay or failure.

Reliability

- It must consistently return correct results and operate under various conditions. **Security**

- • Only authorized admin users can access or modify system data.

- Data should be protected against unauthorized access, corruption, or deletion.

3.4.1 Software Requirements

Component	Specification
Project Title	Rana University Library Management System
Operating System	Windows 7, 8, 10
Backend Technologies	Java (Swing), MySQL Server (via XAMPP)

3.4.3 User Requirements (Functionality)

The primary user is the system administrator. Their expectations include:

- Registering new students and books.
- Searching for issued books or students.
- Updating library records. • Viewing detailed reports.
- Managing overdue penalties and book returns.

3.4.4 Security Requirements

- Every user shall be authenticated with a unique ID and password.
- Only the admin role is granted full control over the system (insert, delete, update).
- Data should be protected from unauthorized modifications.

3.4.5 Performance Requirements

Metric	Requirement
Response Time	Operations should complete within 1 second
Capacity	System should support 10,000+ books records
UI Responsiveness	Screen should respond within 5 seconds
Accessibility	Must conform to Microsoft Accessibility Standards

3.5 Data Gathering Techniques

To ensure the requirements were accurate and representative of real-world needs, multiple data-gathering methods were employed.

3.5.1 Sources of Requirements

- **University Librarian (Primary User)**
- **Library Staff**

- **Supervisor/Advisor Feedback**
- **Observation of Daily Library Operations**
- **Project Documentation from Rana University**

3.5.2 Techniques Used

3.5.2.1 Questionnaire

A set of structured questions was distributed to library staff to understand common challenges with the manual system. This helped in identifying critical features needed in the new system.

3.5.2.2 Interview

Face-to-face interviews were conducted with the university librarian to gather in-depth knowledge about operational workflows, pain points, and desired features.

3.5.2.3 Workshops

An informal discussion session (workshop) was held with the supervising instructor and project advisors to prioritize features and evaluate feasibility.

3.5.2.4 Observation

Real-time observation of the existing manual library process was carried out to understand how books are issued, returned, and tracked.

3.5.2.5 Surveys

Short surveys were administered to gather opinions from students on expected features such as online book availability status and fines.

3.6 Applied Data Gathering Techniques

For the *Rana University Library Management System*, the following methods were used:

- **Observation:** To study and document current manual processes.
- **Interviews:** Conducted with the library administrator for functional requirements.
- **Questionnaires:** Collected responses from stakeholders on system expectations.

This triangulated approach ensured the collection of accurate, validated, and user-centric requirements.

Chapter No 4

System Analysis

System analysis is the process of thoroughly examining existing systems to identify areas for improvement, troubleshoot inefficiencies, and guide the development of new, more effective solutions. In information technology, this process is essential for the successful design and implementation of computer-based systems. It involves analyzing workflows, system architecture, and user interactions to determine how well a system functions and how it can be enhanced.

System analysts play a critical role by evaluating whether a system meets its intended objectives and recommending improvements where necessary. Unlike system administrators who handle daily

operations, analysts take a high-level view to ensure the system aligns with organizational goals and technological best practices.

4.1 Defining the Problem

Defining the problem is a fundamental step in any software development lifecycle. It establishes a clear understanding of the issues that the new system aims to solve and outlines the scope of the project.

In the case of Rana University, the manual library system was inefficient, error-prone, and lacked proper record management. Key issues included:

- Excessive paperwork and redundant data entry.
- Incomplete or inconsistent data.
- High likelihood of data loss or misplacement.
- Difficulty in tracking overdue books and generating reports.

The goal was to transition to an automated Library Management System that minimizes these issues by digitizing operations, improving data accessibility, and providing robust reporting features—while being simple and intuitive for administrative users.

4.2 Preliminary Investigation

Currently, Rana University Library relies on a manual method of record-keeping, which is paper-based and prone to human error. This system lacks proper data centralization and often leads to duplication, data loss, and delays in retrieving critical information.

The limitations of the existing system include:

- Data scattered across various departments.

- Missing or incomplete forms and documentation.
- Inefficient auditing and tracking.
- Multiple versions of the same data leading to inconsistency.

The proposed system aims to resolve these limitations by introducing a centralized, secure, and userfriendly digital system that will enhance the accuracy, speed, and accessibility of library operations.

4.3 Feasibility Study

The feasibility study evaluates whether the proposed system is practical, viable, and worth pursuing from multiple perspectives: technical, economic, operational, and schedule-based.

4.3.1 Technical Feasibility

This involves evaluating whether the available technology and resources can support the proposed system. Key questions include:

- Is the current hardware and software adequate?
- Can the system be developed using existing tools and expertise?
- Will the system be scalable and maintainable?

Technologies Selected

- **Front-End**: Java Swing (NetBeans IDE) ○ Chosen for its GUI capabilities, flexibility, platform independence, and event-driven architecture.
- **Back-End**: MySQL (XAMPP)
    - Selected for its reliability, strong query handling, compatibility with Java, and support for stored procedures.

This configuration ensures that the system is technically feasible and fits within the institutional IT infrastructure.

4.3.2 Economic Feasibility

This assessment examines whether the project can be developed within the financial resources available. Costs considered include:

- Software and hardware requirements.
- Development time and manpower.
- Maintenance and training.

The system is economically feasible, as it leverages free and open-source tools (Java, MySQL, NetBeans, XAMPP), minimizing expenses. Additionally, the increase in efficiency and data reliability is expected to reduce operational costs over time.

4.3.3 Operational Feasibility

Operational feasibility evaluates whether the system can be implemented within the organizational environment and whether users will adopt it.

- Staff training is minimal due to a user-friendly GUI.
- Organizational structure remains largely unchanged.
- The system automates existing workflows, enhancing rather than disrupting operations.

Therefore, the system is considered operationally feasible.

4.3.4 Schedule Feasibility

This checks if the project can be completed within a reasonable time frame. The proposed system had a manageable scope and well-defined requirements, making it feasible to implement within the academic project timeline.

4.4 Project Planning and Scheduling

Effective planning and scheduling are essential to ensure that the project is completed on time and meets all functional and technical requirements.

4.4.1 Development Model: Waterfall

The **Waterfall Model** was adopted for this project due to its structured, linear approach and suitability for well-defined requirements.

Phases:

1. **Requirement Gathering** – Interviews, questionnaires, and observations were conducted.
2. **System Analysis** – Identified problems and defined system goals.
3. **System Design** – Created flowcharts, data structures, and GUI layouts.
4. **Coding** – Implemented the system in Java and MySQL.
5. **Testing** – Validated system functionality.
6. **Deployment** – Delivered the final system for use.

When to Use Waterfall:

- Requirements are stable and well understood.
- Technology stack is established.
- Limited project duration.
- Minimal risk of major changes.

Advantages:

- Easy to understand and manage.
- Clearly defined stages and deliverables.
- Works well for small projects.

Disadvantages:

- Inflexible to requirement changes once development starts.
- No working software until the late stages.
- Poor fit for large or evolving projects.

4.4.2 Project Plan Summary

Phase	Duration
Problem Definition & Analysis	3 weeks
Hardware/Software	

Analysis	2 days
Defining Constraints & Goals	1 week
Requirement Gathering	1 week
Requirement Analysis	1 week
Requirement Specification	2 days
System Design	2 weeks
Coding	2 weeks
Testing and Deployment	3 days

4.5 Risk Management

Risk management helps identify, analyze, and mitigate potential issues that could negatively affect the development or performance of the system.

4.5.1 Risk Identification

Risks were identified in several categories:

- **Future Risks**: Unexpected changes in project direction.
- **Requirement Changes**: Midway modifications in scope.
- **Technical Tools**: Compatibility or skill-based issues.
- **Team Dynamics**: Miscommunication or task overlap.

4.5.2 Risk Analysis

Possible threats include hardware failure, schedule delays, or unexpected requirement changes. Each risk was assessed in terms of its likelihood and potential impact.

4.5.3 Types of Risks

Risk Type	Description
Performance	Risk that the system may not meet all user requirements.
Support	Difficulty in maintaining or enhancing the software.
Cost	Risk of exceeding the planned budget.
Project	Delays or resource issues impacting delivery.
Technical	Technology constraints or integration problems.

Business	Not applicable in this academic context (no commercial risk involved).
Known/Experience	Lack of prior experience as student developers adds uncertainty.

4.5.4 Risk Mitigation Plan

Risk	Solution
Hardware Failure	Regular backups and use of stable hardware.
Schedule Slippage	Clear task allocation and progress tracking.
Changing Requirements	Flexible design and regular stakeholder feedback during early phases.

4.6 Estimation

4.6.1 Effort Estimation

Effort was estimated based on the project size, skill level of developers, and familiarity with tools. The software was designed to be simple, intuitive, and user-friendly for administrators with basic IT knowledge.

4.6.2 Cost Estimation

The project was developed using cost-effective, open-source tools. Key hardware/software requirements:

- **Processor**: Dual-core or higher
- **RAM**: Minimum 512 MB
- **Storage**: At least 60 GB HDD
- **Software**: NetBeans IDE, Java SDK, XAMPP (MySQL)

The development was completed within the budget and resource constraints of an academic project.

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4.4.1 PROJECT DEVELOPMENT MODEL

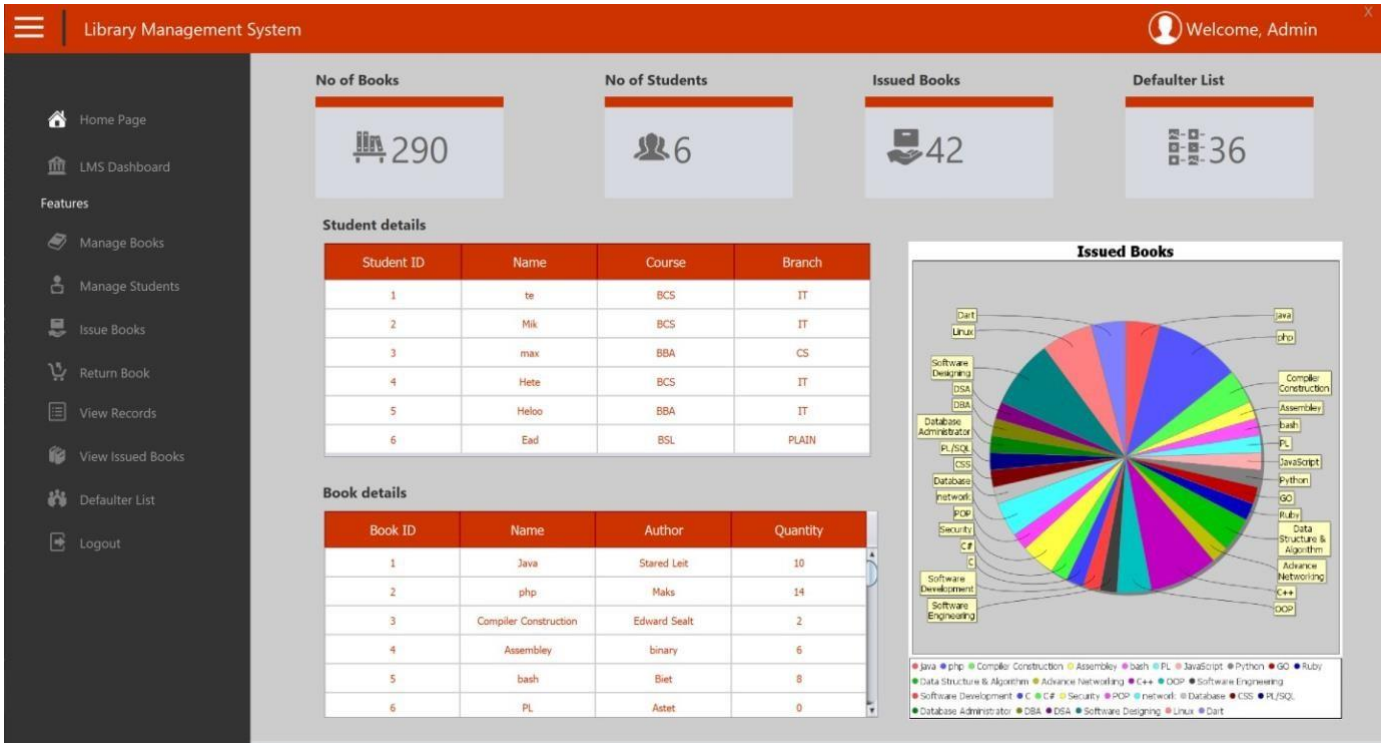

Figure 2 Main Form

The **Waterfall Model** was adopted for this project due to its structured, linear approach and suitability for well-defined requirements.

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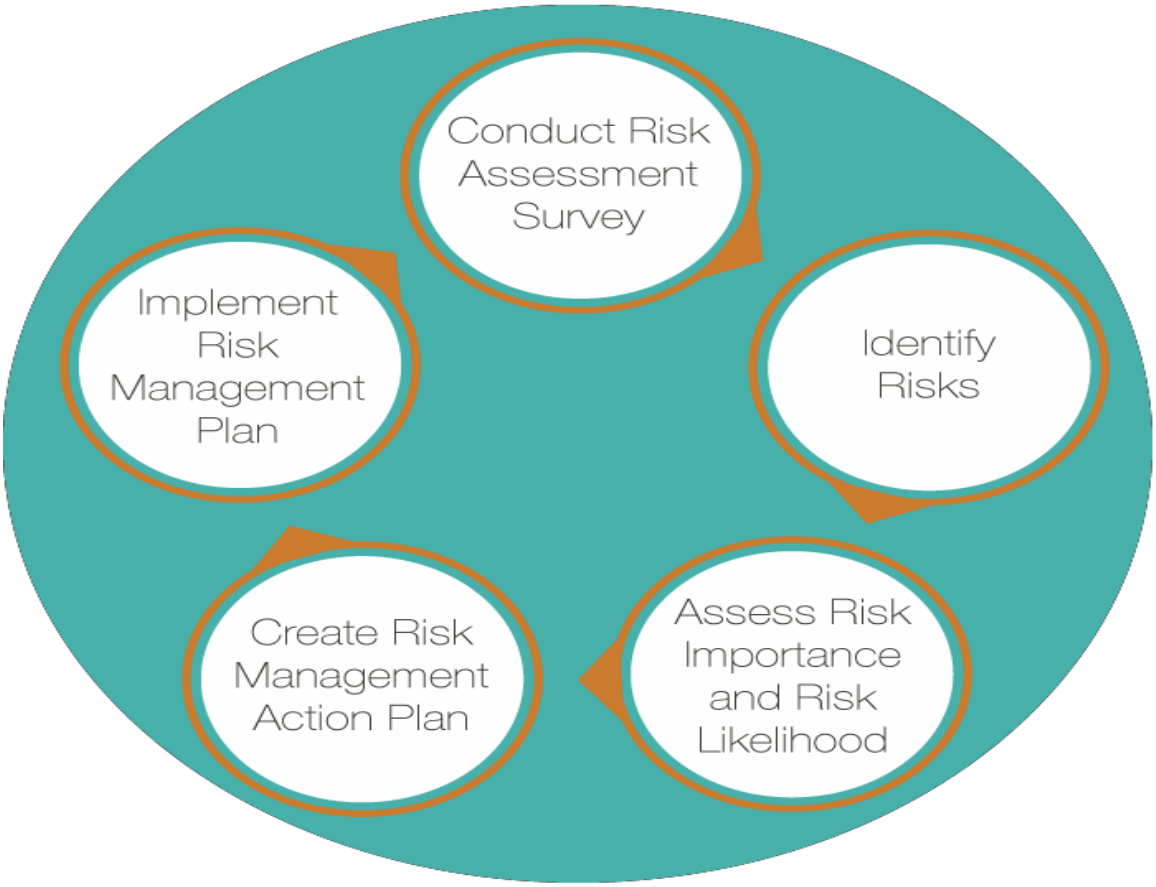

Figure 3 Risk Management

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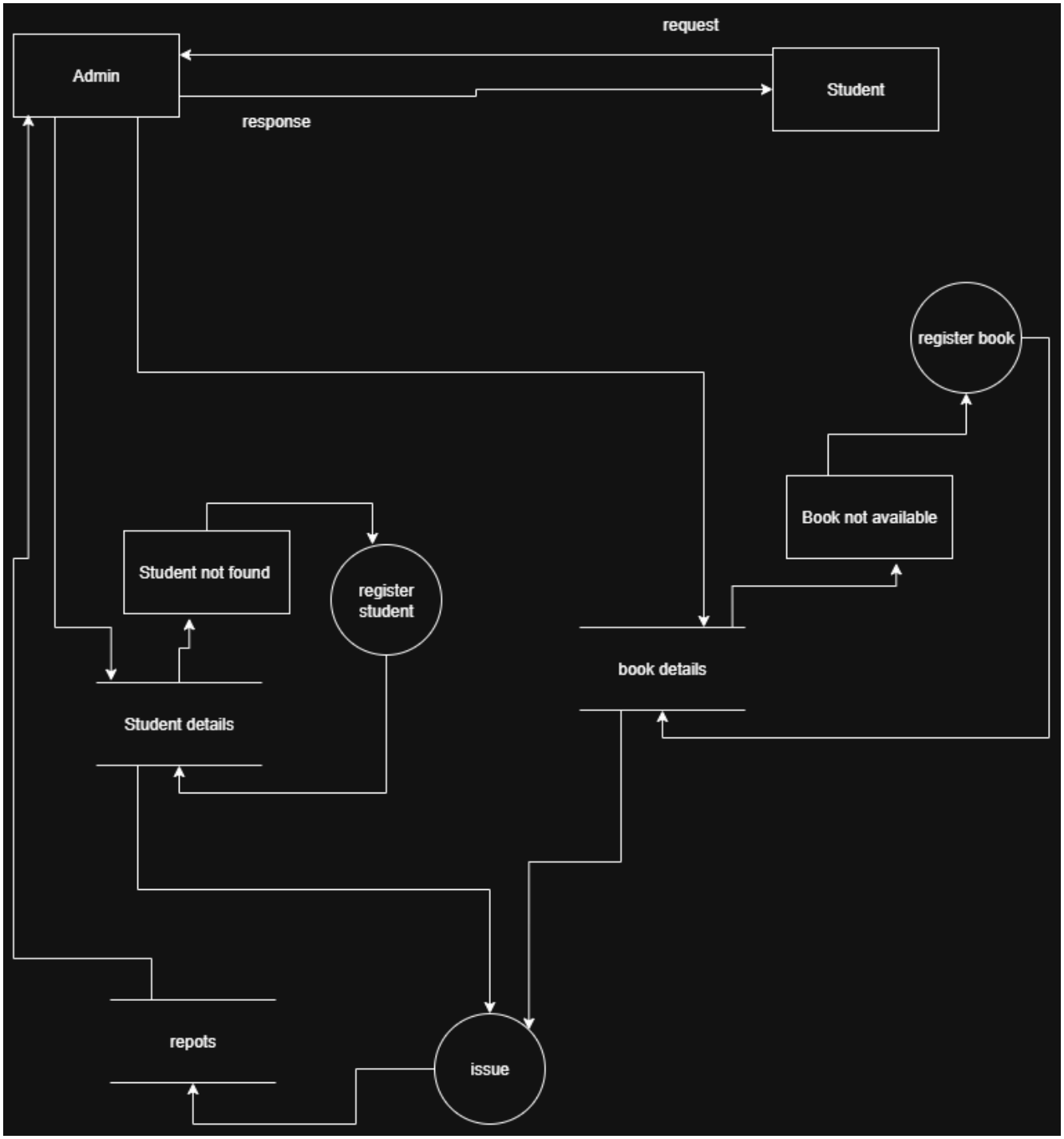


Software for Rana University Library Management System DFD:

Source: MS Word page

4.8 ER Diagram

Entity Relationship diagrams is a specialized graphics that illustrate the interrelationship between entities in a database. ER-modeling is a data modeling technique used in software engineering to produce a conceptual data model of an information system. Diagrams created using this ER-modeling technique are called Entity-Relationship Diagrams, or ER diagrams or ERDs. So you can say that Entity Relationship Diagrams illustrate the logical structure of databases.

There are three basic elements in ER-Diagrams:

- Entities are the "things" for which we want to store information. An entity is a person, place, thing or event.
- Attributes are the data we want to collect for an entity.
- Relationships describe the relations between the entities.

ERDs show entities in a database and relationships between tables within that database. It is essential to have ER-Diagrams if you want to create a good database design. The diagrams help focus on how the database actually works.

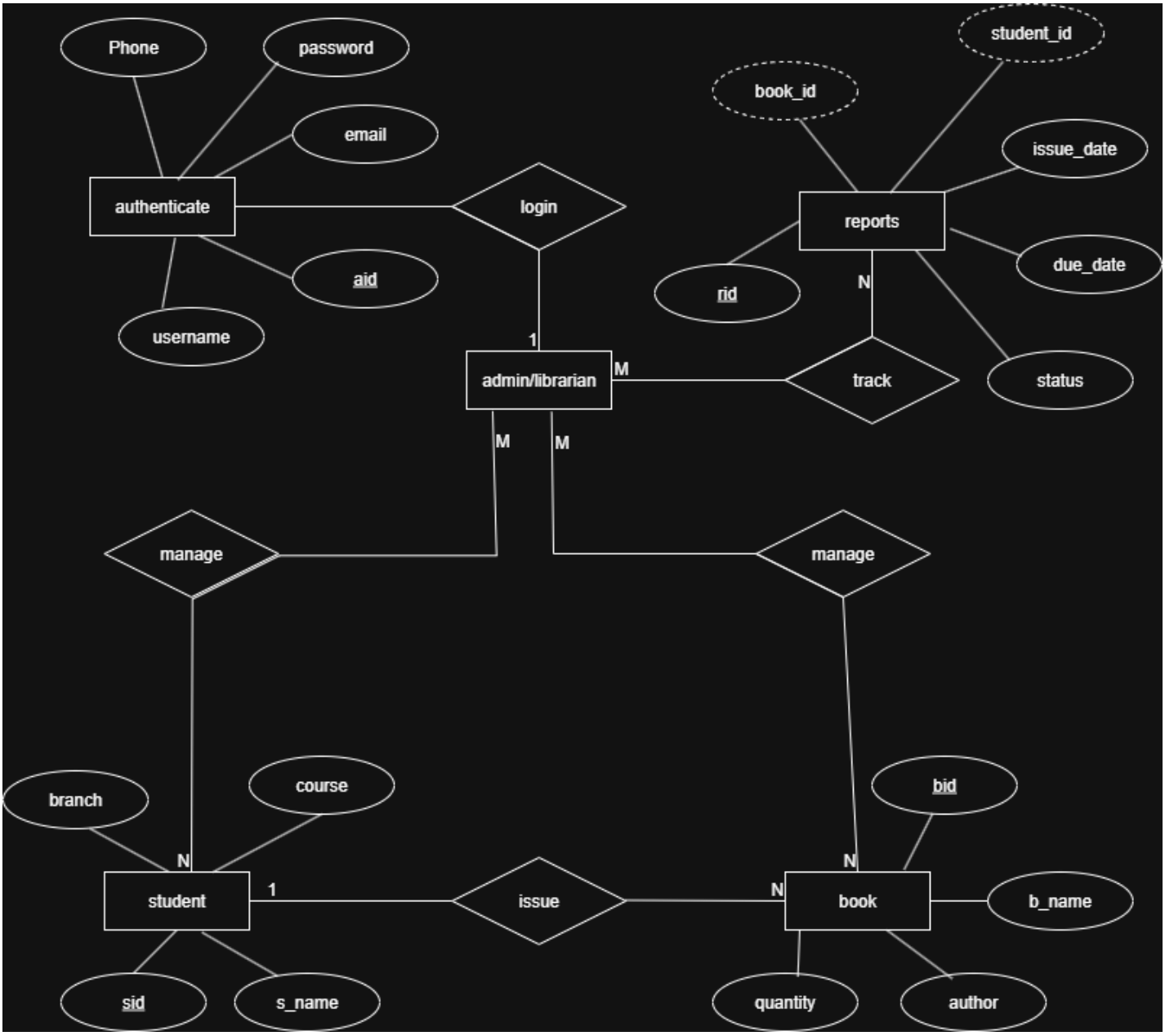


Figure 5 ER Diagram

Source: https://lucidchart.zendesk.com/hc/en-us/articles/360000478183-Template-Entity-RelationshipDiagram

The diagram above is an entity relationship diagram that is a major data modeling tool that helped database analysts to organize data into entities.

4.9 Use case diagram

A UML use case diagram is the primary form of system/software requirements for a new software program underdeveloped. Use cases specify the expected behavior (what), and not the exact method of making it happen (how). Use cases once specified can be denoted both textual and visual representation (such as UML).

A key concept of use case modeling is that it helps us design a system from end user's perspective. It is an effective technique for communicating system behavior in the user's terms by specifying all externally visible system behavior.

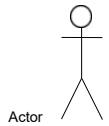
A use case diagram is usually simple. It does not show the detail of the use cases:

- It only summarizes some of the relationships between use cases, actors, and systems.
- It does not show the order in which steps are performed to achieve the goals of each use case.

As said, a use case diagram should be simple and contains only a few shapes. If yours contain more than 20 use cases, you are probably misusing use case diagram.

Use cases represent only functional requirements of a system. Other requirements such as business rules, quality of service requirements, and implementation constraints must be represented separately, again, with other UML diagrams.

4.9.1 Components of Use Case Diagram

Name	Description
	An actor is a direct external user of a system, it is not part of system Actor can be person, device, & other system. Ex. Patient is an actor of hospital management system.
	A use case is a coherent piece of functionality that a system can provide by interacting with actors.
	System boundary is a rectangle in which use case mentioned in & it has system name also. Ex. Hospital Management system.
<<include>>	The include relationship incorporate one use case within the behavior of other use case
<<extend>>	The extend relationship adds increment behavior to a use case.

←	An arrow indicates the relationship among actor & use case.

Table 8 Component of Use Case Diagram

4.9.2 student Use Case Model

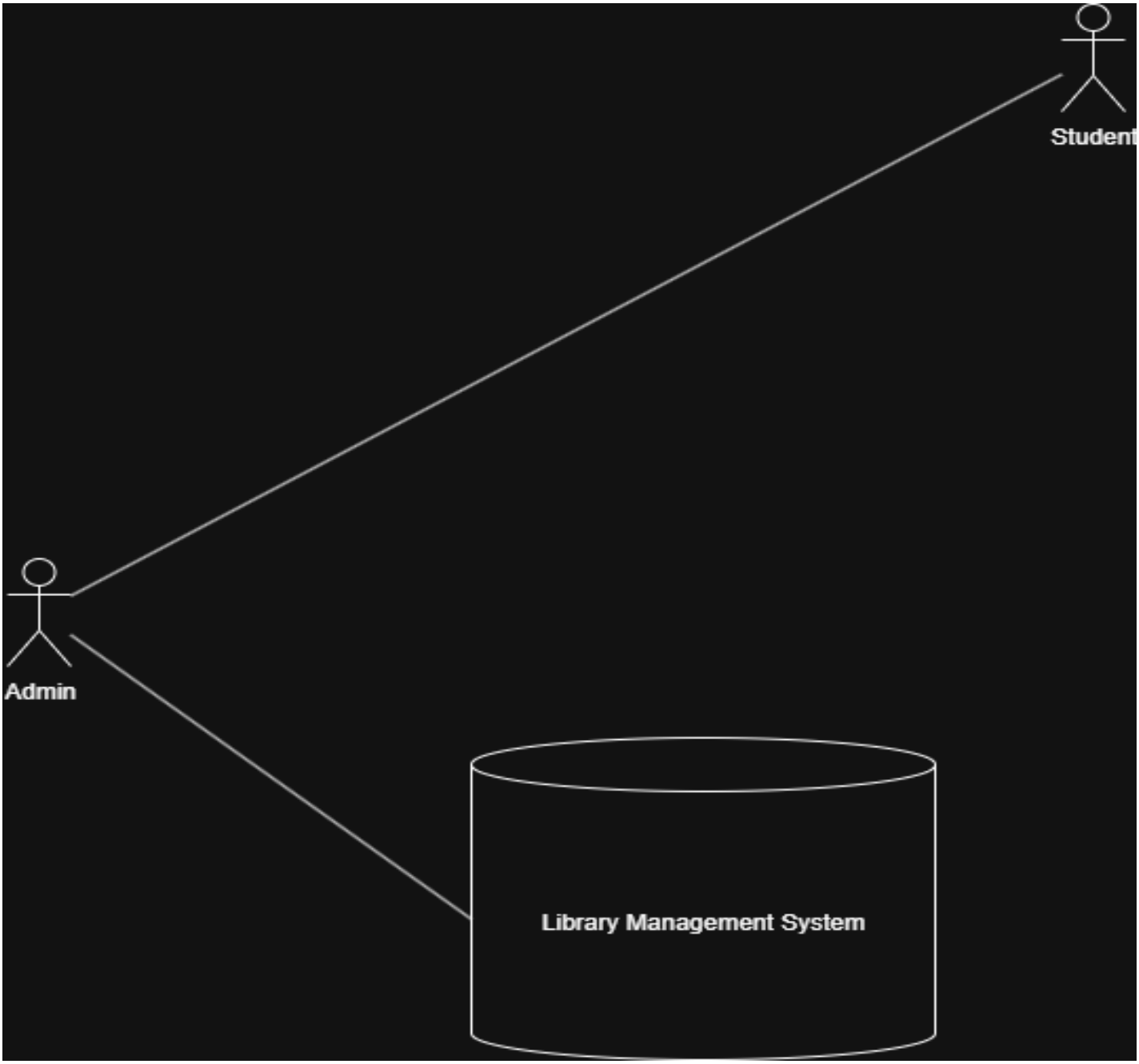
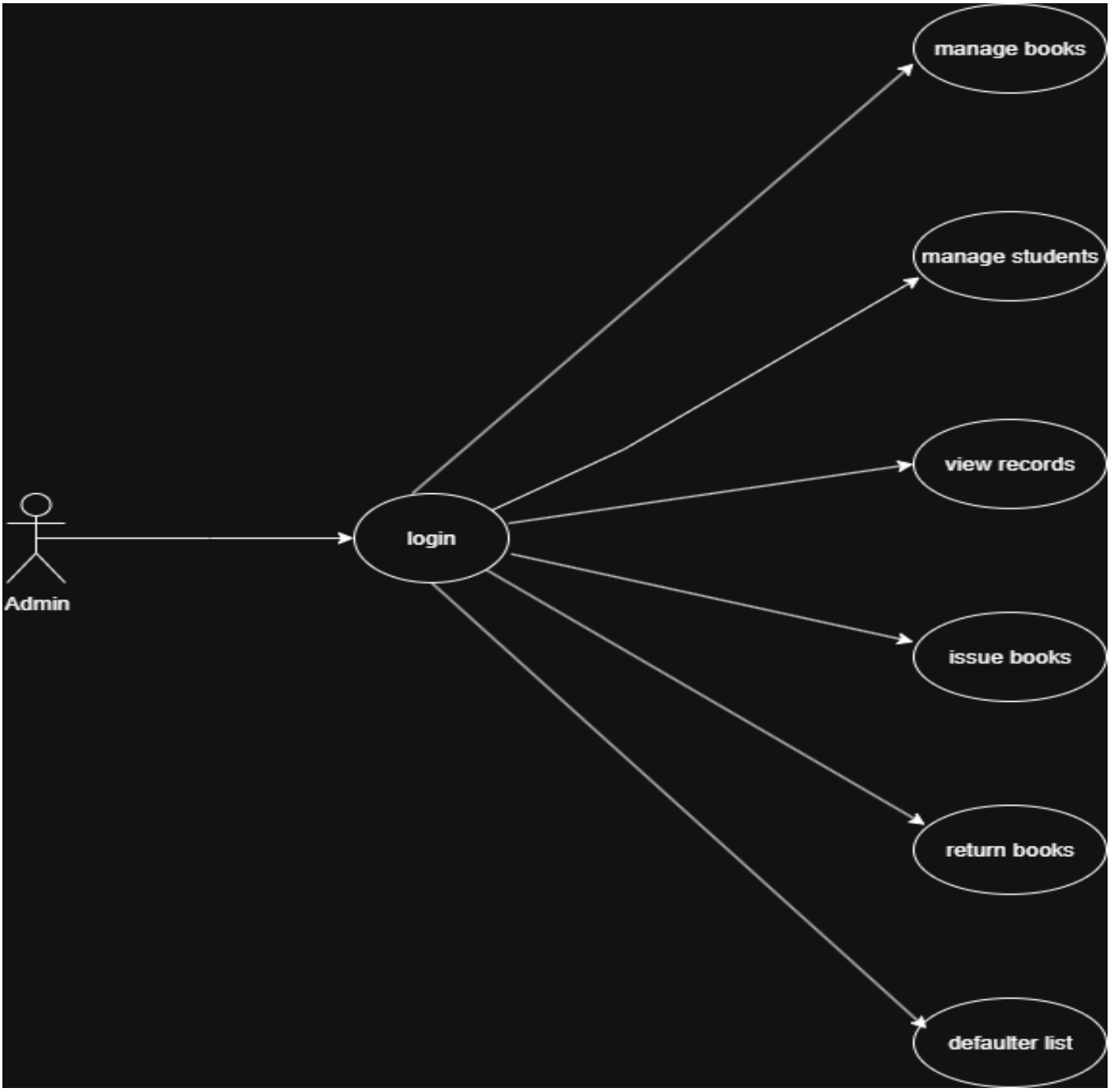


Figure 6 Student use case model

4.9.3 Admin Use Case Model



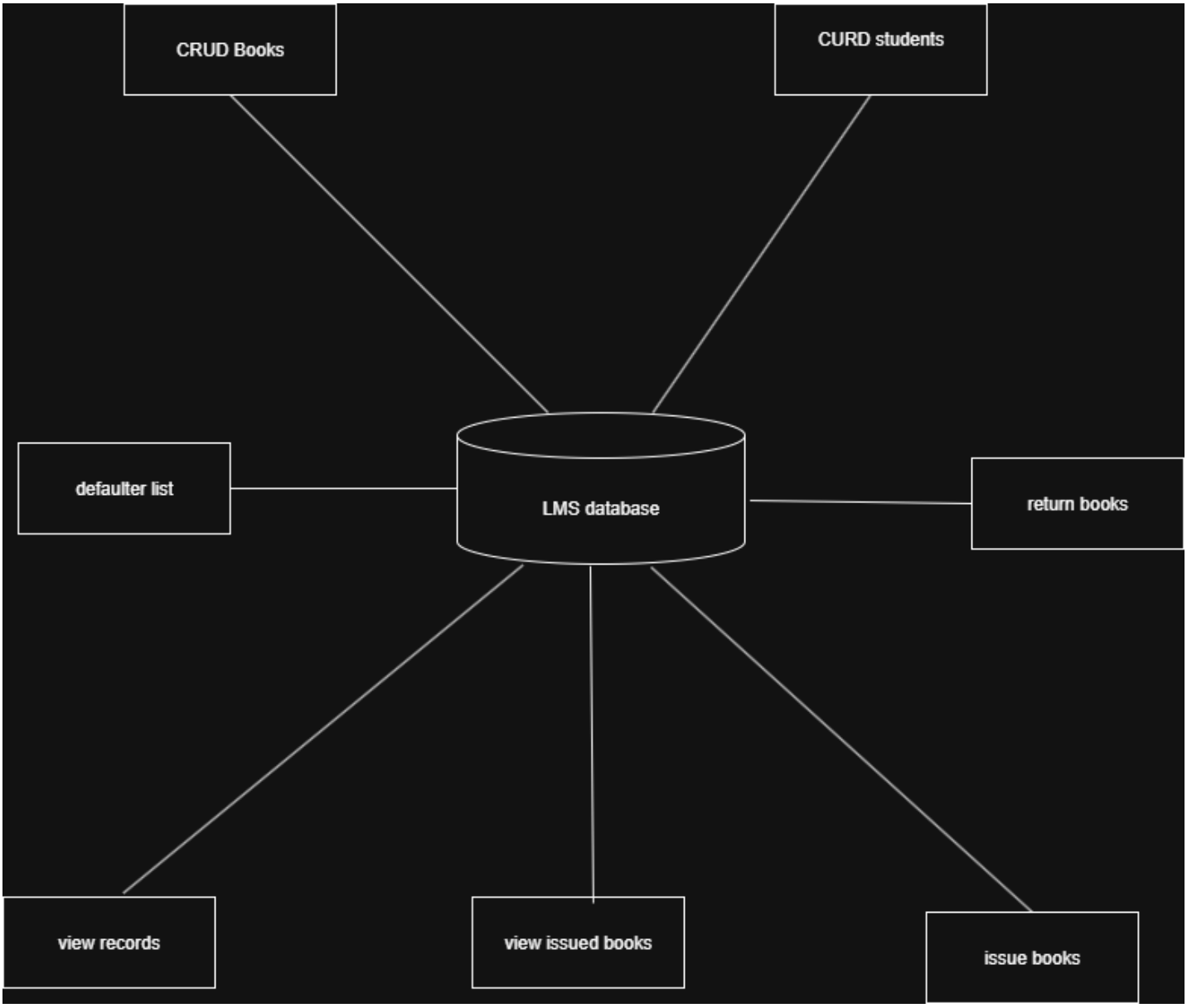

CRUD Books
CURD students
defaulter list
LMS database
return books
view records
view issued books
issue books

Figure 13 Logical Model

Chapter No 5

SYSTEM DESIGN

System design is a crucial phase in the software development lifecycle, as it lays the foundation for building the actual system. This stage involves converting system requirements into a blueprint for implementation, focusing on user interfaces, data management, system architecture, and control

mechanisms. Special attention is given to designing an intuitive and functional user interface, as it significantly affects user experience and system efficiency.

The system design process for the Rana University Library Management System includes the following activities:

- Partitioning the analysis model into functional subsystems.
- Identifying concurrency requirements.
- Allocating tasks and subsystems to appropriate components.
- Designing a user-friendly graphical user interface (GUI).
- Choosing an efficient data management strategy.
- Defining global resources and their control mechanisms.
- Developing task control mechanisms.
- Managing boundary conditions and exceptions.
- Reviewing design alternatives and making informed trade-offs.

5.1 Basic Elements of the User Interface

The user interface (UI) of the Library Management System plays a vital role in facilitating interaction between the user and the system. It is designed using Java Swing components in NetBeans to ensure clarity, ease of use, and responsiveness. The major UI elements include windows, drop-down menus, data entry forms, and control objects.

5.1.1 Windows

A window serves as a graphical container for presenting visual and interactive elements. Multiple windows are incorporated in the system, such as:

- **Login Window** – For secure access to the system.
- **Dashboard/System Console** – For administrative operations.
- **Data Entry Windows** – For adding, editing, or deleting records related to books, students, or transactions.

Each window is designed to be minimalistic, consistent, and task-focused.

5.1.2 Drop-Down Menus

Drop-down menus provide quick navigation across system features. The menu structure is logically grouped, allowing administrators to access modules like:

- Home
- Book Management
- Student Management
- Issue/Return Books
- Records
- Logout

These menus ensure task-oriented navigation, reducing the learning curve for users.

5.1.3 Data Entry Forms

Data entry forms are critical components used to capture information such as student details, book records, and transaction data.

Each form contains:

- **Field Labels (Titles)** – Describe the type of data required (e.g., "Student Name", "Book Title").
- **Field Input Areas (Data Fields)** – Rectangular input boxes where users enter actual data.

Forms are designed to validate inputs to ensure data integrity before submission.

5.1.4 Control Objects

Control objects provide interactive elements that allow users to perform actions. These include:

- **Command Buttons** – Such as "Add", "Update", "Delete", "Clear".
- **Text Boxes** – For entering alphanumeric data.
- **Radio Buttons and Checkboxes** – For selecting options.
- **List Boxes/Drop-down Lists** – For choosing from predefined values.

These controls respond to user actions and trigger appropriate backend processes, such as storing data in the database or retrieving records.

For example, after completing a data entry form, clicking the "Submit" button triggers the event to validate inputs and store the data in the database.

5.2 Modular Design

Modular design enhances system manageability by dividing the system into independent functional units, each responsible for a specific task. This approach improves maintainability, scalability, and testing efficiency.

5.2.1 Project Modules

The Library Management System is composed of the following modules:

1. **Administration**
2. **User Registration**
3. **Book Issue and Return Management**
4. **Book Details Management**
5. **Stock Control and Monitoring**

5.2.2 Module Descriptions

Module Name: Administration

- **Description**: This is the core module of the system and is accessible only by the admin. It controls the entire application, including user access, system settings, and navigation between modules. It acts as the central hub for performing high-level operations and managing the library's digital environment.

Module Name: User Registration

- **Description**: Allows the admin to register students who are eligible to borrow books. Stores student information such as name, ID, contact, and department.

Module Name: Book Issue and Return

- **Description**: Manages the issuing and returning of books. Automatically updates stock quantities and tracks due dates and overdue records.

Module Name: Book Details Management

- **Description**: Stores and manages detailed information about all books in the library, including ISBN, title, author, edition, category, and availability status.

Module Name: Stock Control

- **Description**: Maintains real-time data on available books, issued books, and inventory levels. Helps in identifying shortages or excesses in stock.

5.3 Database Design

A well-structured relational database is central to the operation of the Library Management System. It allows efficient data storage, retrieval, and management while ensuring data consistency and integrity.

5.3.1 Relational Database Management System (RDBMS)

The system uses **MySQL**, a relational database management system, to store and organize data. MySQL is chosen for its reliability, speed, and compatibility with Java applications.

Key features of the database design:

- **Tables** are used to store entities like `Students`, `Books`, `IssuedBooks`, and `Admins`.
- **Primary Keys** uniquely identify each record (e.g., `Student_ID`, `Book_ID`).
- **Foreign Keys** define relationships between tables (e.g., `IssuedBooks` references both `Students` and `Books`).
- **Normalization** ensures that data is logically stored and avoids redundancy.

The database structure is flexible, enabling easy updates, new features, and scalable performance.

5.3.2 Data Dictionary

A data dictionary provides a detailed description of each table, field, data type, and size. It helps in understanding the structure and ensuring consistency during development.
Example snippet:

Table Name	Field Name	Data Type	Description
Students	Student_ID	INT	Unique student identifier
Students	Full_Name	VARCHAR	Name of the student
Books	Book_ID	INT	Unique book identifier
Books	Title	VARCHAR	Book title
Issued Books	Issue_ID	INT	Unique transaction ID
Issued Books	Student_ID	INT	References student borrowing
Issued Books	Book_ID	INT	References book being issued

5.3.3 SQL Integration

Structured Query Language (SQL) is used to create, query, and manage the database. SQL commands are embedded within Java (using JDBC) to enable communication between the application and the MySQL database.

SQL is used for:

- Creating tables and setting constraints.
- Inserting, updating, and deleting records.
- Executing queries to retrieve reports and summaries.
- Managing relationships between entities.

Although SQL is not a general-purpose programming language, it is well-suited for database manipulation. Java, on the other hand, handles the interface, application logic, and user interaction, while SQL executes database transactions.

Figure 14khujasta

5.4 User interface Design

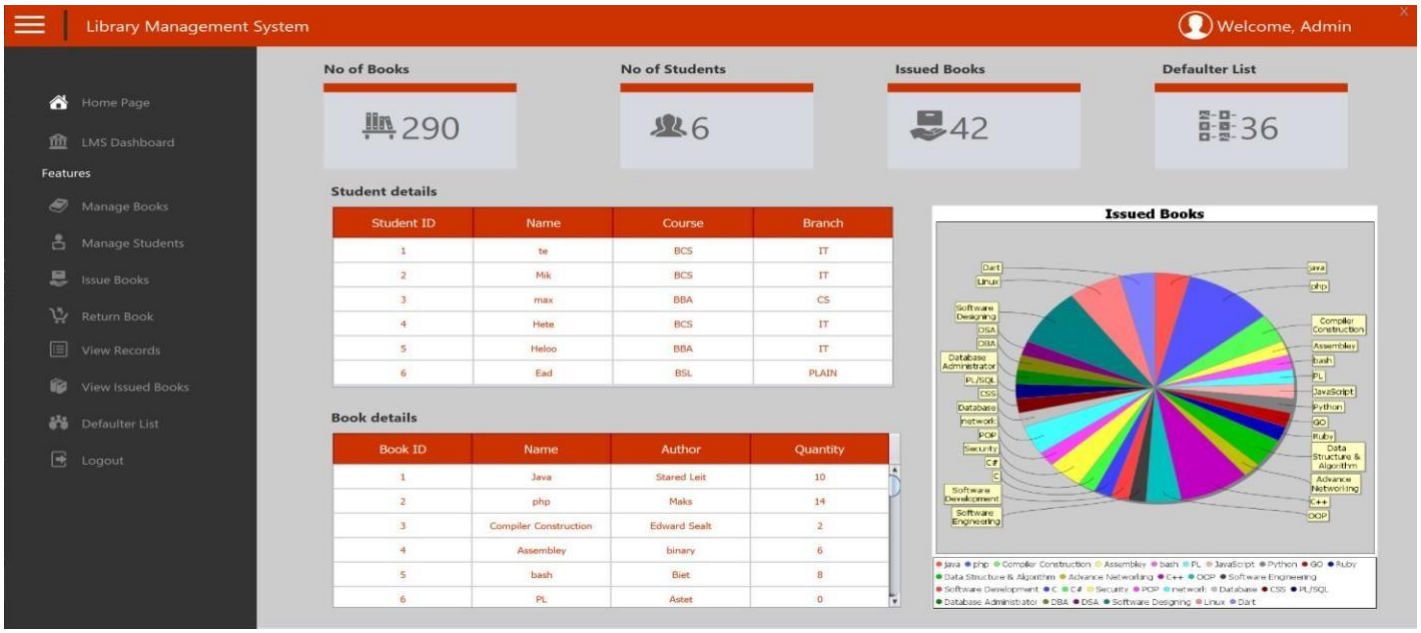


Table 9 User Interface

users

Column Name	Data type	Key
ID	Int	PK
username	varchar(50)	
password	varchar(50)	
email	varchar(50)	
phone	Varchar(50)	

Table 10 Employee

book_details

Column Name	Data type	Key
Book_id	Int	PK
book_name	varchar(50)	
author	varchar(50)	
quantity	int	

Table 11 Customer

Student_details

Column Name	Data type	Key
Student_id	Int	PK
name	Date	
course	varchar(50)	
branch	varchar(50)	

Table 12 Purchase

Issue_book_details

Column Name	Data type	Key
id	Int	PK
book_id	Int	FK
Book_name	varchar(100)	
Student_id	int	FK
Student_name	varchar(50)	
Issue_date	varchar(50)	
Due_date	varchar(50)	
status	Varchar(50)	

Table 13 Stock

5.4 User Interface Design

The system interfaces for the Rana University Library Management System have been developed using **Visual Studio 2015** to ensure a modern and responsive design. For enhanced visual appeal and richer functionality, two third-party component libraries were utilized:

- **DevComponents DotNetBar**: Used primarily for advanced menu bars, toolbars, and navigation controls.
- **Telerik UI for WinForms**: Employed to design interactive and user-friendly controls such as text boxes, buttons, date pickers, and other form elements.

These components significantly improve the look and feel of the application, while also facilitating smoother user interactions.

5.4.1 Login Form

The login form serves as the secure gateway to the system, restricting access to authorized personnel only. The authentication mechanism operates as follows:

- The administrator creates user accounts by defining **username** and **password** credentials, which are securely stored in the database.
- When a user attempts to log in, the system verifies the entered username and password against the stored data.
- Only if the credentials match an existing record will access be granted.
- The login system ensures that unauthorized users cannot access the application, protecting sensitive data and operations.

In this system, the **administrator holds exclusive rights** to access all modules and interfaces.
Additionally, the administrator has the capability to:

- Create new user accounts.
- Manage user permissions and access levels.

This strict control mechanism guarantees that the system remains secure and that user privileges are properly managed.

Figure 15
Login Form

5.4.2 Main Form

The **Main Form** serves as the central hub of the Rana University Library Management System, primarily used by the system administrator to perform key management tasks, including:

- Adding or editing staff and user information
- Managing student records
- Controlling book stock
- Generating various general reports

This main form is designed to be intuitive and accessible, but it assumes that users possess basic knowledge of computer operations and a working understanding of the English language. Without these skills, users may find it challenging to operate the software or navigate the main form effectively.

The Main Form presents all the system modules as menus and command buttons, providing clear and straightforward navigation for the administrator to access different functionalities quickly. Each module is prominently displayed, ensuring that administrative tasks can be executed efficiently from a single interface.

It is important to note that the application is developed using **Visual Basic**, a widely used programming language for standalone desktop applications. Visual Basic allows for the creation of easily deployable packages, which can be installed on any compatible system with minimal effort.

Furthermore, Visual Basic provides a simple yet attractive user interface that simplifies user interaction, making it easier for administrators to manage the application and its data.

Figures 7.2 and 7.3 illustrate the basic main page of the application and the available menu options, respectively. The main screen (Figure 7.2) is consistent across all stores, allowing the user to select the appropriate store and access related operations, as shown in the second screenshot.

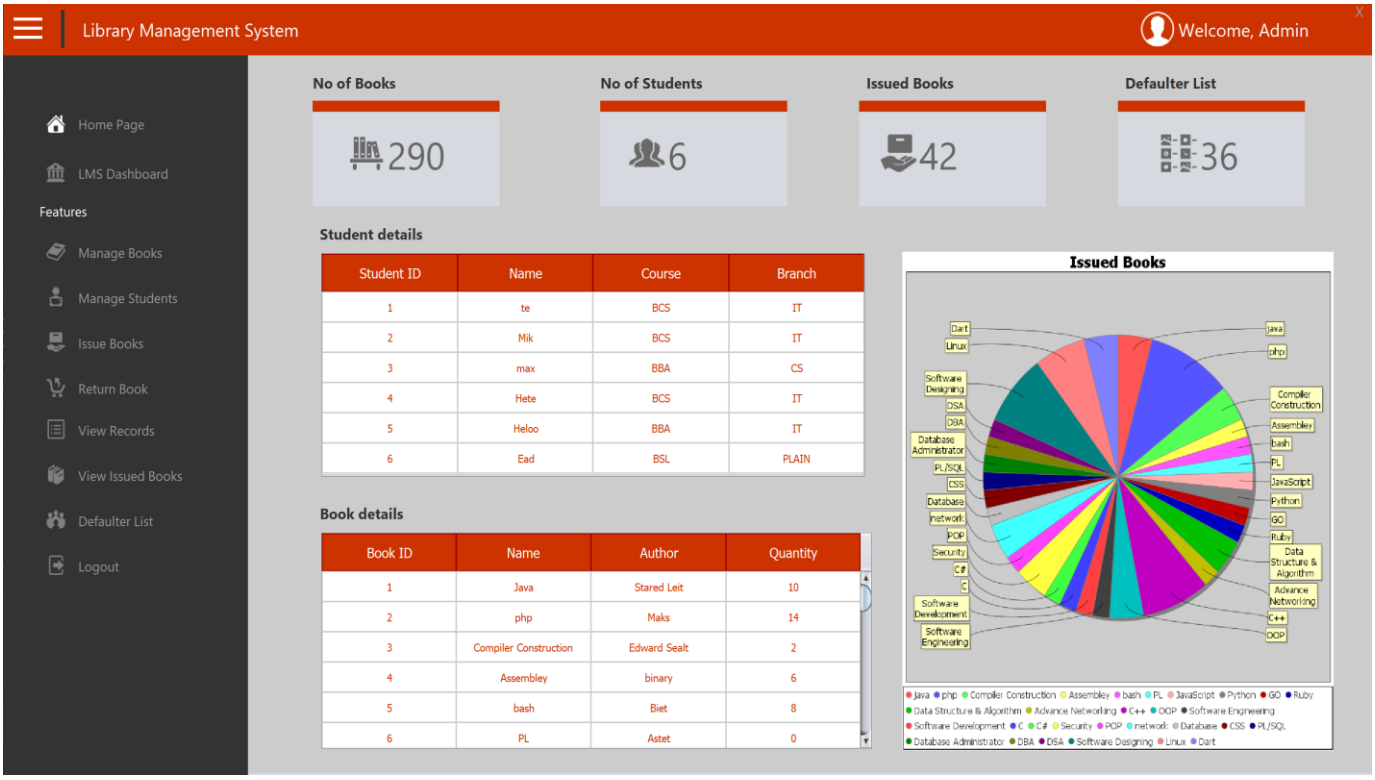


Figure 17 Main Form

Figure 16: Dashboard

The **Dashboard** serves as the startup screen and primary control panel of the Rana University Library Management System. It allows users to access the main features and modules of the system efficiently.

It is important to highlight that users working with this software should have basic computer skills and a functional understanding of the English language. Without these competencies, users may find it difficult to operate the system and utilize the features available through the main form.

From the dashboard, users can select a store (or department) to work with and then access individual forms designed for various specific tasks (as shown in Figure 7.3).

This control panel provides functionality to:

- Select different stores or departments
- Access different types of information related to library operations
- Retrieve overall system information for the Rana University Library Management System

Some of the key options accessible from this dashboard include:

- Add Student
- Add User
- Add Issued Books
- Add New Books
- Validation Control

This design ensures that administrative users can quickly navigate to the necessary modules and perform their tasks without unnecessary complexity, while also maintaining data integrity through validation controls.

5.4.3 Add Student Form:

Student Information is essential to provide enhanced student care facility.

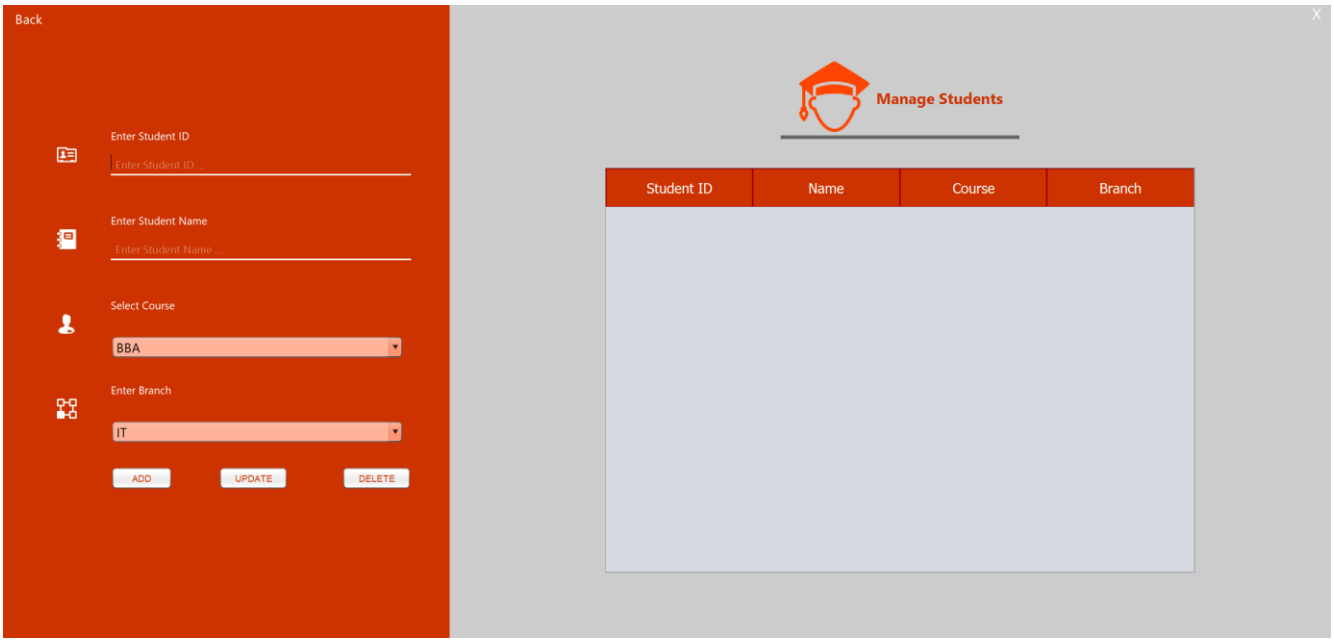


Figure 18 student feedback Form

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Chapter No 6

SYSTEM TESTING AND EVALUATION

Any software cannot be considered complete until it has undergone thorough and repeated testing. Software testing is a major activity that ensures the system is functioning correctly or identifies errors that need to be fixed. The importance of system testing is well summarized by Deutsch:

“The development of the software system involves a series of production activities. Errors may begin to occur at the very inception of the process where objectives may be erroneously or imperfectly specified, as well as during later design and development stages.”

Software testing is a critical component of software quality assurance and plays a vital role in delivering a reliable product.

6.1 TESTING OBJECTIVES

Testing is the process of detecting and correcting as many errors as possible in the developed software. The primary objectives of software testing include:

- **Detecting Errors:** A good test case should have a high probability of discovering previously unidentified errors.
- **Successful Error Discovery:** The success of testing is measured by its ability to uncover errors that have not yet been detected.

These objectives represent a shift from the common misconception that successful testing is merely about systematically covering different types of errors with minimal time and effort. Instead, effective testing focuses on uncovering actual defects.

If testing is conducted effectively, it will reveal errors in the software. As a secondary benefit, it demonstrates that the software functions appear to be working according to the specifications and that performance requirements are met. However, it is important to note that testing cannot prove the absence of errors—it can only confirm their presence.

6.2 SOFTWARE TESTING PRINCIPLES

The following principles guide effective software testing:

- **Traceability:** Tests should be traceable to specific user requirements to ensure that all requirements are verified.
- **Planning:** Testing should be planned well in advance, prior to the actual testing activities.
- **Incremental Testing:** Testing should progress from small-scale tests (unit testing) to larger, more comprehensive testing (integration and system testing).
- **Error Detection:** While testing can confirm the presence of errors, it cannot conclusively prove that the software is error-free.

6.3 SOFTWARE TESTING STRATEGIES

Different testing techniques are appropriate at different stages of the software development lifecycle. Typically:

- Developers conduct initial testing during development.
- For larger projects, an independent testing team is often involved to perform systematic and unbiased testing.

It is important to differentiate between **testing** and **debugging**:

- **Testing** involves executing the software to detect errors.
- **Debugging** is the process of diagnosing and fixing the detected errors.

An effective testing strategy integrates both testing and debugging processes to improve software quality.

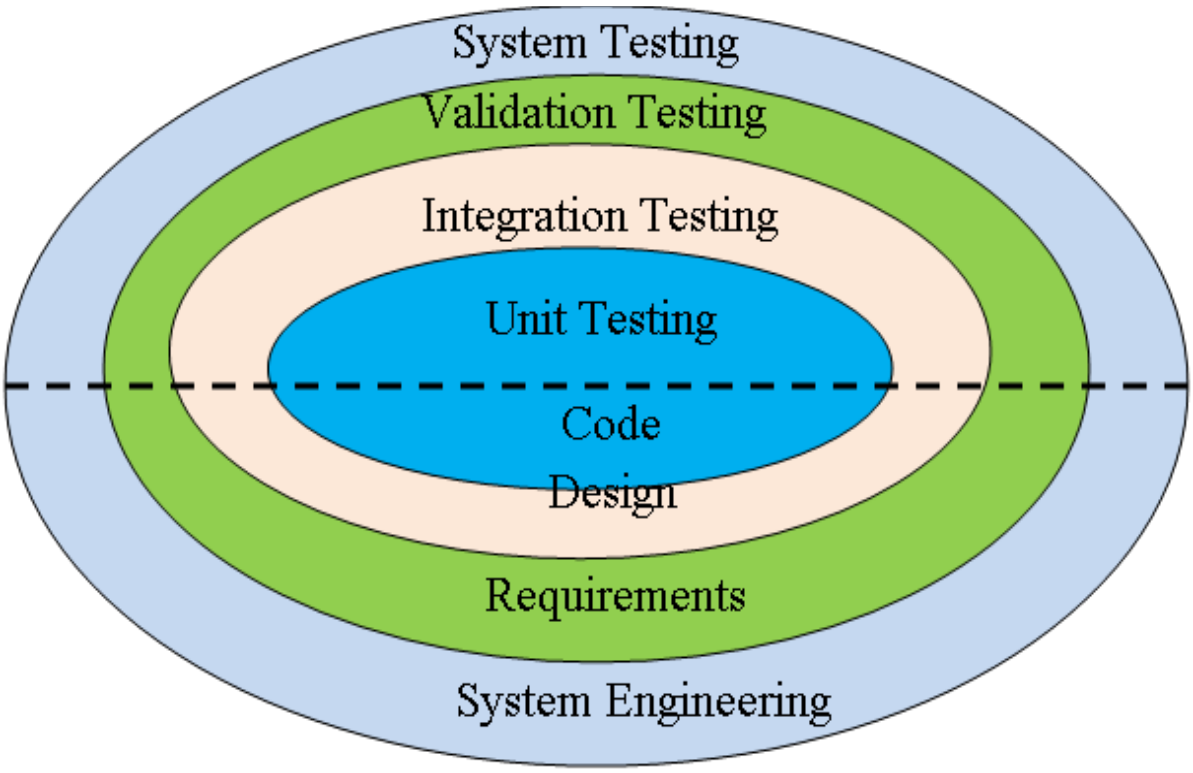

Figure 19 Software Testing Strategy

6.3 SOFTWARE TESTING STRATEGY

A software testing strategy integrates various software test case design methods into a well-planned series of steps, ensuring the successful construction and verification of software. This strategy provides a clear roadmap for software developers, guiding them through the testing process to ensure quality and reliability.

Before discussing specific testing strategies, it is important to understand the different testing techniques that will be used.

6.4 WHAT KIND OF TESTING IS IMPORTANT?

Since the **Rana University Library Management System** is developed with a strong focus on the data tier, specific testing methods must be applied to thoroughly evaluate the application. Generally, two primary testing methodologies are important for database-driven applications: **Unit Testing** and **Functional Testing**.

Unit Testing

Unit testing is a **black box testing** technique that verifies the contracts exposed by the interfaces of individual components or modules. The focus is on testing small, isolated units of code to ensure they produce the correct outputs for given inputs.

- **Input-Output Verification:** For each unit test, specific inputs are provided, and the test verifies if the unit returns the expected output.
- **Pass/Fail Criteria:** A unit test is considered a pass if the output matches the expected valid values in the correct format.
- **Interface Testing:** The procedure or internal workings of the unit are not important for unit tests; what matters is that the interface behaves exactly as specified.

In essence, unit tests ensure that the software components can communicate with the outside world exactly according to their contracts.

Functional Testing

Functional testing is a **white box testing** technique that verifies the overall functionality of the application.

- It involves testing that all parts of the application work correctly together.
- Unlike unit testing, functional testing considers both the internal procedures and the outputs of the application.
- The logic and internal structure of the application are validated, ensuring that all functions behave as expected.

In database applications, functional testing ensures that database operations, business logic, and user interactions perform correctly and produce valid results.

Testing is one of the most important phases in the software development life cycle. It ensures the delivery of high-quality software to end users, minimizing the chances of software issues during actual use. Effective testing validates the software's functionality and provides the following benefits:

- Early detection of defects that can be fixed before the final release.
- Assurance of software stability and reliability under various conditions.
- Delivery of a defect-free software product to the customer.

Testing must be carried out using different methodologies to guarantee comprehensive coverage. Without proper testing, faulty software may be delivered, leading to reduced credibility and dissatisfied users. Typically, testing is conducted by both development and quality assurance teams to validate that the application functions as intended. Since the database is the core module of this system, database testing plays a crucial role in overall system validation.

6.4 SYSTEM TESTING

Once the software has been developed, it is integrated with the entire system—including hardware and other information systems—and subjected to system testing. This phase involves system integration and validation tests that are usually conducted outside the scope of development teams. The end user plays a vital role by identifying errors in the deployed system.

System testing consists of a comprehensive set of tests designed to exercise the complete computer-based system and ensure it operates correctly in its intended environment.

6.5 TESTING SPECIFICATION PLAN

For testing to be effective, it must be consistent and repeatable. Changing test procedures can lead to inconsistent results. Hence, a **test plan** is essential. A test plan is a high-level summary describing:

- The functionalities, components, or regions to be tested.
- The frequency and timing of the tests within the development or deployment process.
- The estimated duration of testing.
- The required resources for testing.

Different phases of software development require different test plans because the testing focus changes over time. Testing newly developed software differs from testing mature software, and any code changes require regression testing.

To decide what to test, the software's purpose, goals, and target users must be clearly understood. Some guiding questions include:

- Why was this software developed?
- What are the business objectives behind it?
- What functionalities are essential for it to work effectively?
- Who are the end users, and can they use the software easily?
- What core features must all users be able to access?

Answers to these questions help define the scope of testing and guide the development of test cases.

6.6 PROJECT TESTING REPORT

The **Rana University Library Management System (RULMS)** has undergone extensive testing, including both black box and white box testing techniques. During the evaluation phase,

the system was reviewed to verify whether its objectives were achieved. A key aspect of this evaluation was feedback from users, as they are the primary operators of the system.

The testing process included:

- Test case descriptions.
- GUI test plans to ensure the interface works as intended.

6.7 LIMITATIONS

The Rana University Library Management System has several limitations due to time constraints and initial design decisions:

- Users must have basic computer skills to operate the system effectively.
- The system currently allows entry of only one item at a time; bulk data entry is not supported.
- There is only one shared username and password, meaning all users share the same credentials.
- The search functionality is limited to searching by ID number only, requiring users to know the exact ID of the entity they wish to find.

6.8 FUTURE EXPANSION

Future enhancements for the system may include:

- Support for multiple usernames and passwords with role-based access control.
- Expanded search functionality allowing queries by various criteria.
- Integration of employee timecards and payroll management.
- Implementation of SIM card management.

- Use of barcode scanners and smart card readers to reduce manual data entry and improve efficiency—though these additions will increase system cost.

Conclusion

The **Rana University Library Management System** is designed to computerize daily library management activities efficiently. It provides effective storage and management of information related to book registration, issuance, and student records.

Developing this software was challenging due to the complexities of software engineering, requiring significant effort in coding, research, and testing. The main goal was to deliver a comprehensive system for Rana University, and although only key modules are covered in this thesis due to contractual limitations, the core objectives were achieved.

At the outset, the project appeared straightforward but revealed increasing complexity as development progressed. While the current system is standalone and lacks communication with other computers in different branches, future versions can address this limitation.

Currently, the system experiences some performance delays due to technical issues that could not be resolved within the thesis timeline. However, the project remains ongoing, with continuous bug fixes and improvements planned.

Ultimately, this project successfully delivered a functional system for managing book registration and issuance at Rana University, laying the foundation for future enhancements and broader deployment.

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